

量子 2016

- 一、1. 能量交换光谱 (吸收 → 发射)
2. 能量交换光谱.
3. 粒子的波动性.
4. 光场的粒子性
5. 粒子的波动性. (电子衍射)

二、费米子：交换反对称

$$\psi = \psi_n(x_1) \psi_n(x_2) \cdot \frac{1}{\sqrt{2}} [x_{1\uparrow} x_{2\downarrow} - x_{1\downarrow} x_{2\uparrow}]$$

↓ 合并 $m=n$

$$\psi = \frac{1}{2} [\psi_n(x_1) \psi_m(x_2) + \psi_m(x_2) \psi_n(x_1)].$$

($m \neq n$)

$$[x_{1\uparrow} x_{2\downarrow} - x_{1\downarrow} x_{2\uparrow}]$$

$$\psi = \frac{1}{\sqrt{2}} [\psi_n(x_1) \psi_m(x_2) - \psi_m(x_2) \psi_n(x_1)].$$

($m \neq n$)

$$\begin{cases} x_{1\uparrow} x_{2\uparrow} \\ \frac{1}{\sqrt{2}} [x_{1\uparrow} x_{2\downarrow} + x_{1\downarrow} x_{2\uparrow}] \\ x_{1\downarrow} x_{2\downarrow} \end{cases}$$

玻色子：交换对称

$$\psi = \psi_n(x_1) \psi_n(x_2) \begin{cases} x_{1\uparrow} x_{2\uparrow} \\ \frac{1}{\sqrt{2}} [x_{1\uparrow} x_{2\downarrow} + x_{1\downarrow} x_{2\uparrow}] \\ x_{1\downarrow} x_{2\downarrow} \end{cases}$$

↓ 合并 $m=n$

$$\psi = \frac{1}{\sqrt{2}} [\psi_n(x_1) \psi_m(x_2) + \psi_m(x_1) \psi_n(x_2)].$$

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$$\begin{cases} x_{1\uparrow} x_{2\uparrow} \\ \frac{1}{\sqrt{2}} [x_{1\uparrow} x_{2\downarrow} + x_{1\downarrow} x_{2\uparrow}] \\ x_{1\downarrow} x_{2\downarrow} \end{cases}$$

$$\psi = \frac{1}{2} [\psi_n(x_1) \psi_m(x_2) - \psi_m(x_1) \psi_n(x_2)].$$

($m \neq n$)

$$[x_{1\uparrow} x_{2\downarrow} - x_{1\downarrow} x_{2\uparrow}]$$

$$\text{三、(1)} \quad P = |\psi(r, \theta, \varphi, t)|^2 \cdot 4\pi r dr$$

$$(2) \quad P = \left[\int_0^\infty |\psi(r, \theta, \varphi, t)|^2 r^2 dr \right] d\Omega$$

四、(1) ψ_1, ψ_2, ψ_3 是 $\hat{H}_0$ 的本征波函数。
在 $\hat{H}_0$ 表象中，取 ψ_1, ψ_2, ψ_3 为基。

$$\hat{H}_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \hbar\omega & 0 \\ 0 & 0 & 2\hbar\omega \end{pmatrix}$$

$$\psi_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \psi_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \psi_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

(2) 矩阵元

$$H'_{mn} = \langle \psi_m | H' | \psi_n \rangle, \quad m, n = 1, 2, 3$$

成立条件 $\lambda \ll 1$

五、(1) 在 S_z 表象中， $\psi_0 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

由 $\hat{S}_x \psi = S_x \psi$ 得出

$$S_{x+} = \frac{\hbar}{2}, \quad \psi_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$S_{x-} = -\frac{\hbar}{2}, \quad \psi_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

可能测量值为 $\pm \frac{\hbar}{2}$

$$(2) \quad P(S_x = \frac{\hbar}{2}) = |\langle \psi_+ | \psi_0 \rangle|^2 = \frac{1}{2}$$

$$P(S_x = -\frac{\hbar}{2}) = |\langle \psi_- | \psi_0 \rangle|^2 = \frac{1}{2}$$

$$(3) \quad \vec{n}(\theta) = \sin\theta \cos\varphi \vec{i} + \sin\theta \sin\varphi \vec{j} + \cos\theta \vec{k}$$

$$S_0 = \vec{n}(\theta) \cdot \vec{S} = \sin\theta \cos\varphi S_x + \sin\theta \sin\varphi S_y + \cos\theta S_z$$

$$= \frac{\hbar}{2} \begin{pmatrix} \cos\theta & \sin\theta e^{-i\varphi} \\ \sin\theta e^{i\varphi} & -\cos\theta \end{pmatrix}$$

$$\text{由 } \frac{\hbar}{2} \begin{pmatrix} \cos\theta & \sin\theta e^{-i\varphi} \\ \sin\theta e^{i\varphi} & -\cos\theta \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = E \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$E_+ = \frac{\hbar}{2}, \quad \psi_+ = \begin{pmatrix} \cos\frac{\theta}{2} e^{-i\frac{\varphi}{2}} \\ \sin\frac{\theta}{2} e^{i\frac{\varphi}{2}} \end{pmatrix}$$

$$E_- = -\frac{\hbar}{2}, \quad \psi_- = \begin{pmatrix} \sin\frac{\theta}{2} e^{-i\frac{\varphi}{2}} \\ -\cos\frac{\theta}{2} e^{i\frac{\varphi}{2}} \end{pmatrix}$$

可能取值为 $\pm \frac{\hbar}{2}$

$$(4) \quad \langle S_0 \rangle = \langle \psi_0 | S_0 | \psi_0 \rangle$$

$$= (0, 1) \frac{\hbar}{2} \begin{pmatrix} \cos\theta & \sin\theta e^{-i\varphi} \\ \sin\theta e^{i\varphi} & -\cos\theta \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= -\frac{\hbar}{2} \cos\theta$$

六. 和2018年第五题一样.

七. 微扰 H'

$$|1\rangle = |u_0\rangle, \quad |2\rangle = |u_M\rangle, \quad |3\rangle = |u_2\rangle$$

$$\langle 1 | H' | 1 \rangle = \langle 2 | H' | 2 \rangle = \langle 3 | H' | 3 \rangle = 0$$

$$\langle 1 | H' | 2 \rangle = \langle 1 | H' | 3 \rangle = \langle 2 | H' | 3 \rangle = \hbar\omega_1$$

$$H' = \hbar\omega_1 \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$\text{由 } \begin{vmatrix} E - \hbar\omega_1 & -\hbar\omega_1 & -\hbar\omega_1 \\ -\hbar\omega_1 & E & -\hbar\omega_1 \\ -\hbar\omega_1 & -\hbar\omega_1 & E \end{vmatrix} = 0$$

$$\Rightarrow (E + \hbar\omega_1)^2 (E - 2\hbar\omega_1) = 0$$

$$E_1 = 2\hbar\omega_1, \quad E_2 = -\hbar\omega_1 \quad (= \text{重简并})$$

$$E_1 = 2\hbar\omega_1, \quad \psi_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$E_2 = -\hbar\omega_1, \quad \psi_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \psi_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$t=0 \text{ 时, } \psi_0 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{3}} \psi_1 + \frac{\sqrt{2}}{3} \psi_2 - \frac{\sqrt{2}}{3} \psi_3$$

$$= \frac{1}{\sqrt{3}} \psi_1 + \frac{1}{\sqrt{2}} \psi_2 + \frac{1}{\sqrt{6}} \psi_3$$

$$\psi(t) = \frac{1}{\sqrt{3}} e^{-iE_1 t/\hbar} |\psi_1\rangle + \frac{1}{\sqrt{2}} e^{-iE_2 t/\hbar} |\psi_2\rangle + \frac{1}{\sqrt{6}} e^{-iE_3 t/\hbar} |\psi_3\rangle$$

$$= \frac{1}{3} e^{-i2\omega_1 t} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \frac{1}{2} e^{i\omega_1 t} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \frac{1}{6} e^{i\omega_1 t} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$= \frac{1}{3} \begin{pmatrix} e^{-i2\omega_1 t} + e^{i\omega_1 t} \\ e^{-i2\omega_1 t} - e^{i\omega_1 t} \\ e^{-i2\omega_1 t} + e^{i\omega_1 t} \end{pmatrix}$$

$$P(u_M) = \left| \frac{1}{3} (e^{-i2\omega_1 t} - e^{i\omega_1 t}) \right|^2 = \frac{2}{9} (1 - \cos 3\omega_1 t)$$

$$P(u_2) = \left| \frac{1}{3} (e^{-i2\omega_1 t} + e^{i\omega_1 t}) \right|^2 = \frac{2}{9} (1 - \cos \omega_1 t)$$